WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

The Cost of the Prompt: Prompt-Free SAM 2 on Sentinel-2 Fields

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MOTIVATION & RESEARCH PROBLEM

Meta SAM 2

- SAM 2 [1] segments anything from **one human click**. Mapping thousands of fields needs it **prompt-free**: the automatic generator clicks a regular point grid instead.
- Sentinel-2 parcels are far from SA-1B [2] objects: 10 m pixels, a median parcel of **113 px (≈ 11 × 11 px)**.

Question: how well does frozen, prompt-free SAM 2 find agricultural parcels, and what limits it: the model or the prompt?

OBJECTIVES / CONTRIBUTION

- **Teardown** of frozen SAM 2.1: parameters, compute per block, latency.
- **Baseline** of the automatic generator against parcel ground truth.
- **Diagnose** the prompt: grid density and SAM's fixed mask filters.
- **Adapt** the mask decoder (encoder frozen) and test it with the deployment pipeline.

METHODOLOGY

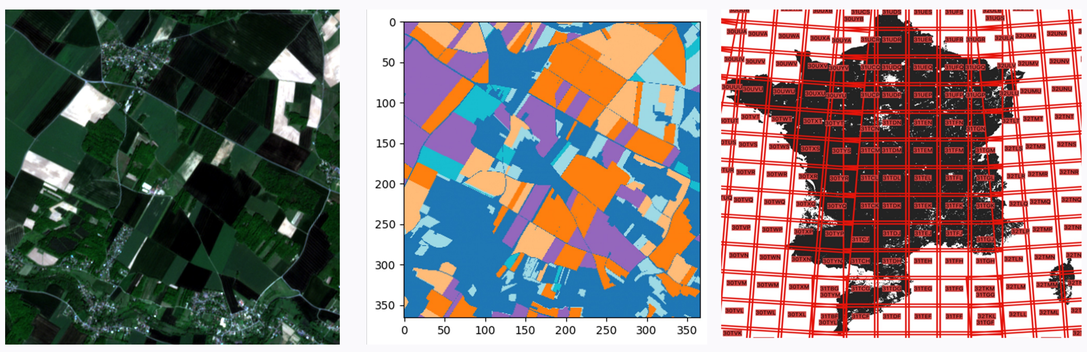


Fig. 1. A test patch (RGB) and its parcel map; the 6 Sentinel-2 tiles lie in Catalonia and France.

- **Data:** Sen4AgriNet [3], 211,375 parcels (≥ 20 px) in 888 Sentinel-2 patches (366 × 366 px, RGB, 10 m, May), 6 tiles, Spain and France. Split 70/15/15 by patch, stratified by tile, seed 42; test = 132 patches.
- **Model:** SAM 2.1 Hiera-Small; automatic generator, 16 × 16 grid (pps = 16), filters from SA-1B: predicted IoU ≥ 0.5, stability ≥ 0.8.
- **Metric:** for each true parcel, best IoU over all generated masks; mean over parcels; paired bootstrap 95 % CI.
- **Adaptation:** train decoder transformer layer 2 + IoU head (1.71 M params, 3.7 %), Dice (+ IoU-head) loss. v1: centroid prompts. v2: prompt (centroid / random / grid) × output tokens, checkpoint chosen by generator IoU on val.
- **Pre-registered:** expectations written before each run (D = random prompts predicted best; E_noiou won).

CONCLUSIONS

- **The cost of the prompt** is three things: grid density (32% of parcels reached at pps 16), filters calibrated on another domain (off: 0.156 → 0.262), and the train/inference prompt mismatch (v1: 0.064).
- **Do not retrain the IoU head under fixed thresholds**, or recalibrate the thresholds: with its loss off, the decoder beats frozen SAM (0.156 → 0.242).
- **Validate on the deployment pipeline:** proxy gains (val prompted IoU) predicted the opposite of test.

Limits: one seed; 150 training patches; pps 64 on 40 Spanish patches only; timings on a shared GPU.

KEY REFERENCES

[1] N. Ravi et al., “SAM 2: Segment Anything in Images and Videos,” arXiv:2408.00714, 2024.
[2] A. Kirillov et al., “Segment Anything,” in Proc. IEEE/CVF ICCV, pp. 4015–4026, 2023.
[3] D. Sykas et al., “A Sentinel-2 Multiyear, Multicountry Benchmark Dataset for Crop Classification and Segmentation With Deep Learning,” IEEE JSTARS, vol. 15, pp. 2937–2953, 2022.

RESULTS & DISCUSSION

A. Frozen baseline and cost. The encoder costs 134 GMAC per image, the decoder 1.4 GMAC per prompt (16 × 16 grid = 2.6 × the encoder). Automatic mode on test: mean best IoU **0.156** (132 patches, pps 16); Spain 0.085 ≪ France 0.220.

B. Prompt density. A 16 × 16 grid reaches only **32%** of parcels (32 × 32: 68%, 64 × 64: 95%). pps 32 on all 132 test patches: **0.253** (Δ +0.096 [+0.088, +0.104]) at 5–9 × the time; pps 64: 0.185 vs 0.083 on 40 Spanish patches.

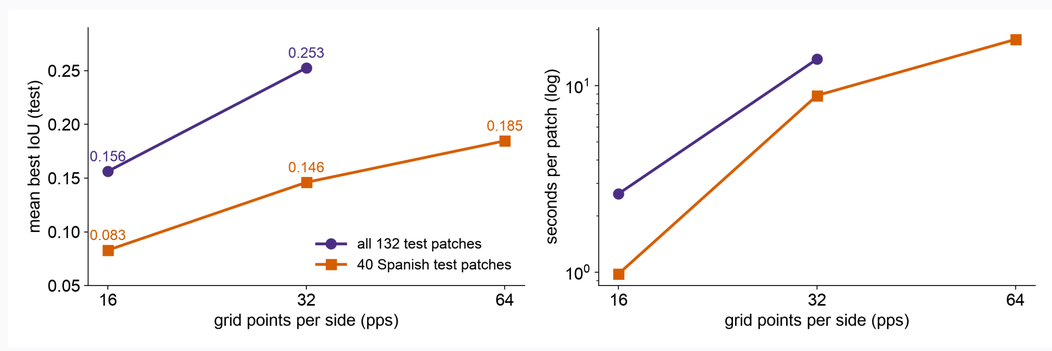


Fig. 2. Test mean best IoU (left) and seconds per patch (right, log; shared pod GPU) vs. grid density, on all 132 test patches (pps 16/32) and 40 Spanish test patches (pps 16/32/64).

C. SAM's filters, calibrated on SA-1B, discard good masks. Filters off (pps 16): **0.262** (Δ +0.106 [+0.100, +0.112]), 329 vs 79 masks per patch; share of mask area on fields 0.58 → 0.47: a precision trade-off.

D. Fine-tuning. v1 (centroid prompts) raised prompted mask IoU on val 0.312 → 0.530 but the generator on test collapsed to **0.064**. v2: training the IoU head lowers the generator metric every time (selection kept epoch 0) although masks improve (filters off: E_loss 0.304 vs 0.262). **IoU-head loss off** (grid prompts, tokens 1–3): test **0.242**, Δ +0.085 [+0.080, +0.091], the first fine-tune that beats frozen SAM. Negative clicks + step-level selection: 0.213.

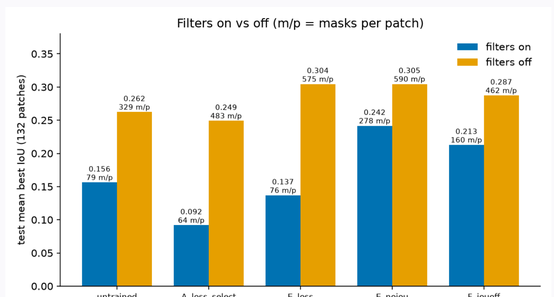


Fig. 3. Test mean best IoU with SAM's filters on / off (m/p = masks per patch).

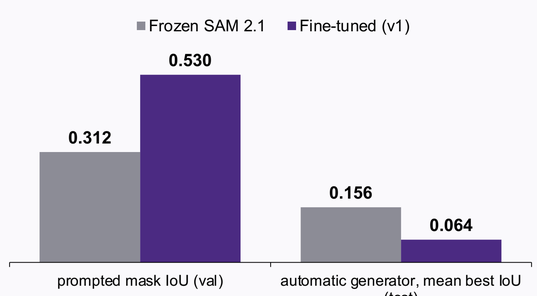


Fig. 4. v1: val bars = prompted mask IoU; test bars = automatic generator, mean best IoU.

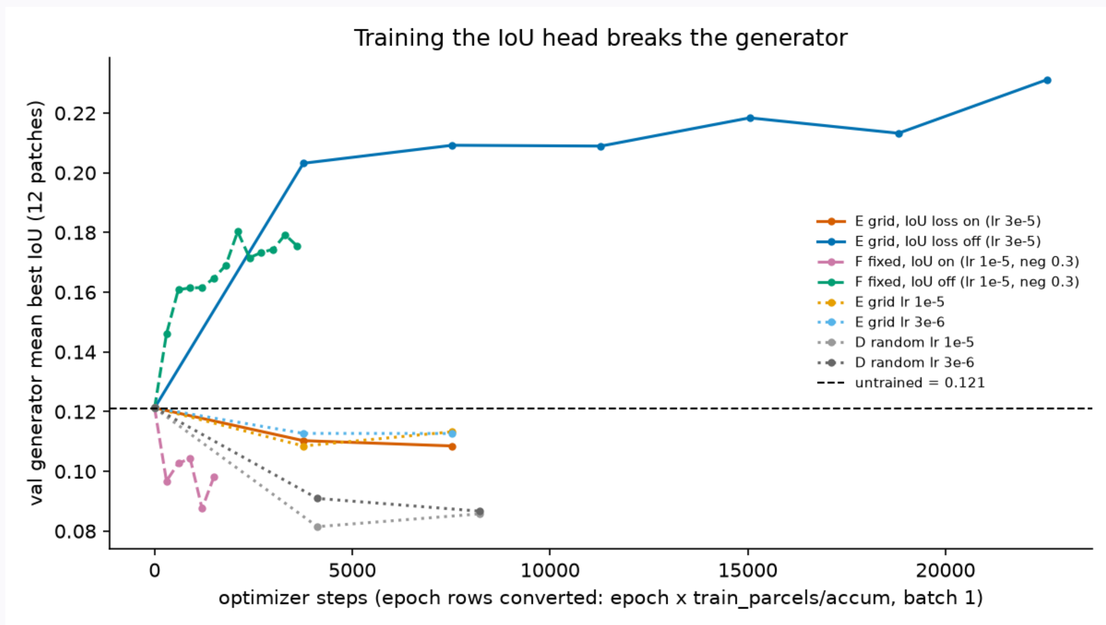


Fig. 5. Validation generator IoU during training (12 val patches). Every run that trains the IoU head falls below the frozen model; with the IoU-head loss off it rises.

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CONTINUE THE CONVERSATION

Contact: andre.rubio@pucp.edu.pe
Code, splits and results:
github.com/ZagonTM/Sen4Sam

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